

ECE 3441 Spring 2023

Exam 1

Version A

Please do not open booklet unless instructed to do so, or substantial penalty will result.

- Closed book, closed notes
- ONE 8.5 x 11" double-sided cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 80 minutes.

First Name Last Name,

Peoplesoft ID,

Date *DD/MM/YYYY*

Question	Points	Maximum
1		15
2		15
3		20
4		20
5		25
6		20

Total: _____/100 possible points + 10 Bonus

Question 1:

- Convert 110010100.0111_2 to base 8.
- Do the following 10-bit 2s complement binary addition and indicate overflow/no overflow:

$$\begin{array}{r} 0110101101 \\ + 1000101110 \\ \hline \end{array}$$

a)

$$\underbrace{110}_6 \underbrace{010}_2 \underbrace{100}_4 . \underbrace{011}_3 \underbrace{100}_4$$

$$(6 \times 8^2) + (2 \times 8^1) + (4 \times 8^0) + (3 \times 8^{-1}) + (4 \times 8^{-2})$$

$$384 + 16 + 4 + 0.375 + 0.0625$$

404.4375

b)

$$\begin{array}{r} 111111 \\ 0110101101 \\ + 1000101110 \\ \hline \end{array}$$

111101101₂

No - overflow

Question 2:

Find the minimum SOP (three terms) of the following using Boolean algebra:

$$F_{2A}(A, B, C) = (AC \oplus B') + (A'B + B'C)'$$

State any theorems used in the simplification process (*Hint: consensus theorem must be used to get the minimum SOP, but it is not the only theorem/law used in the process*).

You must use Boolean algebra to receive any credit.

$$\begin{aligned} F_{2A}(A, B, C) &= (AC)'B' + AC(B')' + (A'B)'(B'C)', \quad \text{XOR, DeMorgan} \\ &= (A' + C')B' + ACB' + (A + B')(B + C)', \quad \text{DeMorgan} \\ &= \underbrace{A'B'}_{yz} + \underbrace{B'C'}_{yz} + ACB' + \underbrace{AB + AC'}_{xy}, \quad \text{10L, 1L} \end{aligned}$$

Question 3:

Find the **minimum** sum-of-products expression for the following maxterm expansion using a 5-variable Karnaugh map. (There is only one correct solution)

$$F_{3A}(A, B, C, D, E) = \prod M(0, 1, 4, 5, 8, 12, 16, 17, 20, 21, 24, 26, 27, 28)$$

(Hint: Use the minterms instead of the maxterms when plotting the k-map. The final answer has 4 terms)

$A = 0$

$BC \backslash DE$	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	1	1	1	1
10	1	1	1	1

$$F_{3A} = A'DE' + A'BE + A'B'D + BDE + ABD + ACD$$

$A = 1$

$BC \backslash DE$	00	01	11	10
00	0	0	0	0
01	0	0	1	1
11	1	1	1	0
10	1	1	1	0

Question 4:

Determine the prime implicants of the following minterm expansion using the Quine-McCluskey method.

$$F_{4A}(A, B, C) = \sum m(0, 1, 2, 4, 5, 6, 7)$$

(Hint: 3 columns are required and the final answer is 2 terms.)

0	0000 ✓	0,1	000- ✓	0,2,4,6	0--0
1	0001 ✓	0,2	00-0 ✓	0,4,2,6	0--0
2	0010 ✓	0,4	0-00 ✓	0,4,1,5	0-0-
4	0100 ✓	1,5	0-01 ✓	0,1,4,5	0-0-
5	0101 ✓	4,5	010- ✓	4,5,6,7	01--
6	0110 ✓	2,6	0-10 ✓	4,6,5,7	01--
7	0111 ✓	4,6	01-0 ✓		
		5,7	01-1 ✓		
		6,7	011- ✓		

prime implicants:

$A'D'$
 $A'C'$
 $A'B$

Question 5:

Eliminate any static 0 hazards in the function below and write the new equation in POS form.

$$F(A, B, C, D) = (B + D')(C' + D)(A + B' + C)$$

(Hint: Look at the 0s on the k-map)

		AB			
CD		00	01	11	10
		1	0	1	1
01		0	0	1	0
11		0	1	1	0
10		0	0	0	0

$$F(A, B, C, D) = (B + D')(C' + D)(A + B' + C)(A + C + D')(A + B + C')(A' + B + C')$$

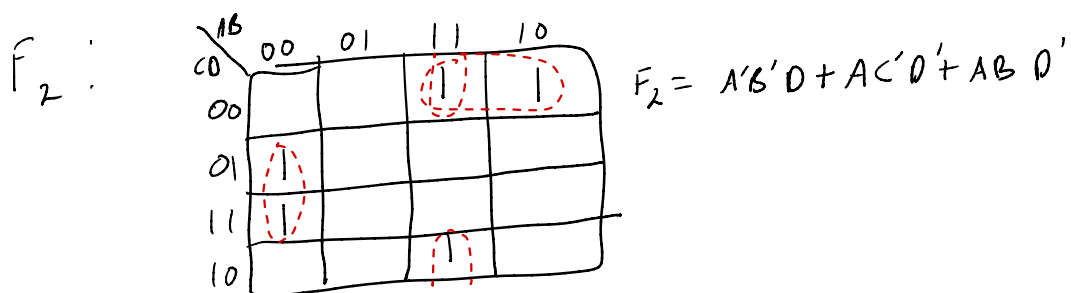
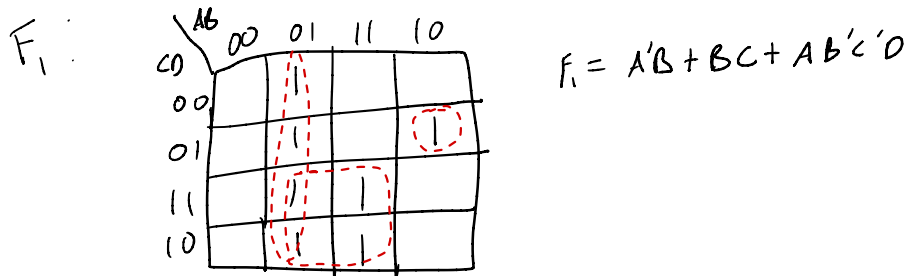
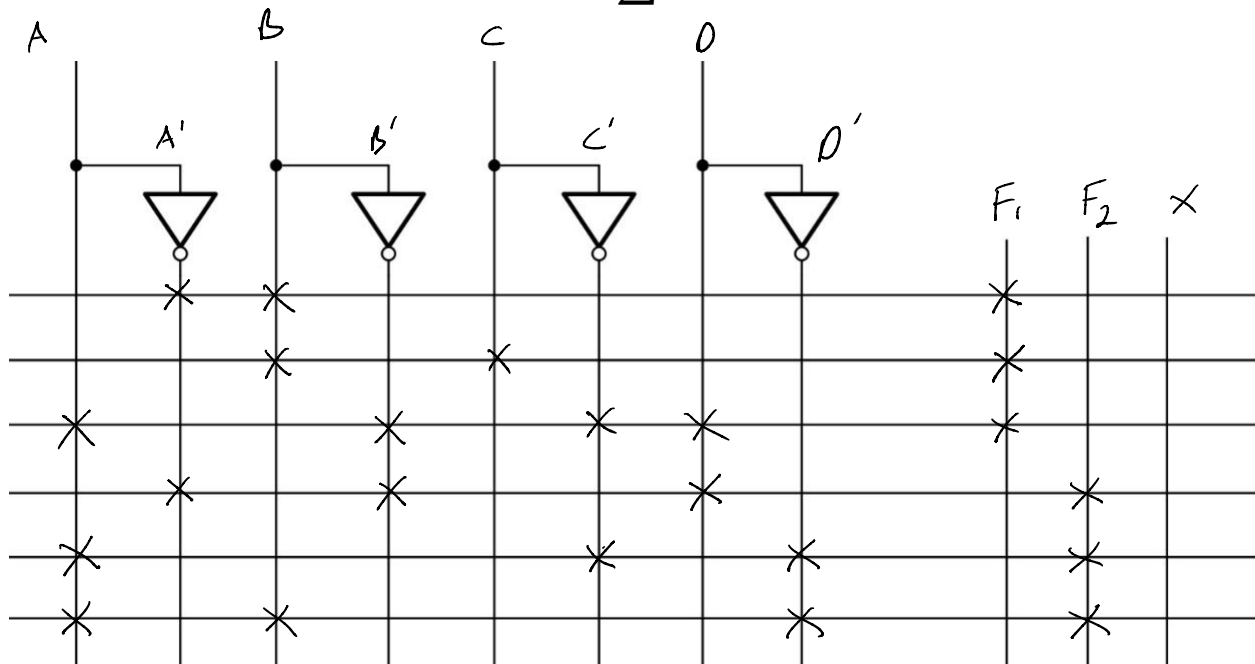
Question 6:

Implement the following 3 functions using the blank PLA table provided.

(Hint: you will need to simplify each function to a minimum SOP in order for all 3 functions to fit.)

$$F_1(A, B, C, D) = \sum m(4, 5, 6, 7, 9, 14, 15)$$

$$F_2(A, B, C, D) = \sum m(1, 3, 8, 12, 14)$$



Extra Paper